which leads to the probability density

$$\rho_{\mathrm{NPT}} = Z_{\mathrm{NPT}}^{-1} e^{-\beta(H+\mathrm{PV})},$$

where H is the Hamiltonian, and $\beta^{-1} = \mathrm{k_B T}$.

The NPT model may be viewed as corresponding to a thermostatted system which incorporates a "piston" controlling the overall volume of the system to be consistent with the target pressure. During simulation, the volume must be free to change; whereas the methods presented so far retain a fixed volume. In NVE or NVT simulation, the volume enters in two ways into the molecular simulation method: (i) as a modification of the potential (allowing for periodic replicas) and (ii) in the mechanism by which atoms are mapped from one boundary to the other of the rectangular domain, simulating motion on a torus. It is not obvious how to generalize these to the NPT setting where V is a dynamical variable.

The method of Andersen [11] ("Andersen's piston") is to control the volume by reparamaterizing the coordinates to lie in a cubic box with unit side. Assuming the system occupies a cubic domain, say, $[-L/2, L/2] \times [-L/2, L/2] \times [-L/2, L/2]$, where $L = V^{1/3}$, we may write, for each three-dimensional atomic coordinate vector $\boldsymbol{q}_i$

$$\boldsymbol{q}_i = V^{1/3} \bar{\boldsymbol{q}}_i, \tag{8.71}$$

where $\bar{\boldsymbol{q}}_i$ is confined to the unit-side cube. Then we have

$$\dot{\boldsymbol{q}}_i = \frac{1}{3} V^{-2/3} \dot{V} \bar{\boldsymbol{q}}_i + V^{1/3} \frac{\mathrm{d}}{\mathrm{d}t} \bar{\boldsymbol{q}}_i.$$

The kinetic energy of the atomic system may be written

$$K = \sum_{i=1}^{N} m_i \frac{\|\dot{\boldsymbol{q}}_i\|^2}{2}.$$

After coordinate transformation this becomes

$$\begin{aligned} \bar{K} &= \sum_{i=1}^{N} \frac{m_i}{2} \left\| \frac{1}{3} V^{-2/3} \dot{V} \bar{\boldsymbol{q}}_i + V^{1/3} \frac{\mathrm{d}}{\mathrm{d}t} \bar{\boldsymbol{q}}_i \right\|^2 \\ &= \sum_{i=1}^{N} \frac{m_i}{2} \left[\frac{1}{9} V^{-4/3} \dot{V}^2 \left\| \bar{\boldsymbol{q}}_i \right\|^2 + \frac{2}{3} V^{-1/3} \dot{V} \bar{\boldsymbol{q}}_i \cdot \dot{\bar{\boldsymbol{q}}}_i + V^{2/3} \left\| \dot{\bar{\boldsymbol{q}}}_i \right\|^2 \right], \end{aligned}$$

thus the momentum vector conjugate to $\bar{\boldsymbol{q}}_i$ becomes

$$\bar{\boldsymbol{p}}_i = \frac{\partial \bar{K}}{\partial \dot{\bar{\boldsymbol{q}}}_i}$$